

Problems marked with a (*) are a bit more complex.

5.1. Derivatives.

- (a) Consider $f(x, y) := xy^2e^{-x^2-y^2}$ for $(x, y) \in \mathbb{R}^2$. Find all the critical points of f , that is all the points where the gradient of f vanishes. (The point will be given if and only if all the numerical values you find are correct... so check your computations twice).
- (b) We want to find a function $g \in C^1(\mathbb{R}^2)$ with the following directional derivative:

$$\partial_v g(x, y) = 2 \cos(x^2 y) x v_2 + \cos(x^2 y) v_1^2 y \quad \text{for all } (x, y) \text{ and } (v_1, v_2) \in \mathbb{R}^2.$$

Give an explicit example of such a function g or prove that a function with these properties cannot exist.

5.2. Lagrange multipliers. Consider the ball $U := \{(x, y, z) : x^2 + y^2 + z^2 < 1\}$ and the function

$$f(x, y, z) := 1 - x^2 - y^2 + 4x.$$

1. Motivate rigorously why f attains its absolute maximum and minimum in $\overline{U}$.
2. Find all the critical points of f which lie in the interior of U . Say whether $\sup_U f$ (or $\inf_U f$) is attained at some point in U .
3. Say whether $\max_{\partial U} f$ and $\min_{\partial U} f$ exist.
4. With the method of Lagrange multipliers, find the possible critical points for the constrained problem

$$\max\{f(x, y, z) : (x, y, z) \in \partial U\},$$

and same for min.

5. Among all the points you have found clarify at which points the maximum/minimum of f is attained.

5.3. Multiple choice. Mark all and only the statements which are true

- (a) As $|x| \rightarrow 0$, if $f(x) = x_2 x_1^2 + O(x_1^2)$, then $f(x) = O(|x|^2)$.
- (b) As $x \rightarrow 0$, if $f(x) = x_2 x_1^2 + O(x_1^3)$, then $f(x) = O(x_1^3)$.
- (c) As $|x| \rightarrow 0$, if $f(x) = x_2 x_1 + O(|x|^2)$, then f is differentiable at $x = 0$.
- (d) As $x \rightarrow 0$, if $f(x) = x_2 x_1 + O(|x|^2)$, then f is twice differentiable around $x = 0$.
- (e) As $|x| \rightarrow 0$, if $f(x) = x_2 x_1 + O(|x|^3)$, then f is twice differentiable around $x = 0$ and $\partial_{11} f(0) = 0$.

5.4. Computation of derivatives. For each of the following functions compute the directional derivative in the general direction v at a general point x , that is $\partial_v f(x)$, where applicable.

1. $x_1/x_2, e^{-x_1/x_2}$ for $x \in \mathbb{R}^2 \setminus \{x_2 = 0\}$. (Also: Can they be extended continuously or differentiably across $\{x_2 = 0\}$?)
2. $e^{x_1 x_2} \sin(x_1 + x_2^2), \frac{x_1^2}{x_1^2 + x_2^2}, \frac{x_1^2 x_2}{x_1^2 + x_2^2}, \frac{x_1^2 x_2^2}{x_1^2 + x_2^2}$ for $x \in \mathbb{R}^2 \setminus \{0\}$. (Also: Can they be extended continuously or differentiably across the origin?)
3. $|x|, |x|^\alpha, x/|x|, g(|x|), x \cdot e, g(x \cdot e)$, for $\alpha \in \mathbb{R}, e \in \mathbb{R}^n$ and $g \in C^1(\mathbb{R})$.
4. $ax, x^T, \text{Tr}(x), x^2, x^3, \text{Tr}(x^2)$ for $x, a \in \mathbb{R}^{n \times n}$,
5. (*) x^{-1}, x^{-2} for $x \in \mathbb{R}^{n \times n}$ with $\det x \neq 0$.

5.5. Multiple choice. Assume $g \in C^\infty(\mathbb{R})$ and $f \in C^\infty(\mathbb{R}^n)$ is such that

$$f(x) = x_1 + x_2 x_1 + O(|x|^4) \text{ as } x \rightarrow 0.$$

We want to compute the Taylor expansion of $(g \circ f)$ at $x = 0$ of the highest possible order with the information we have.

We can ask for the value $g^{(k)}(t)$ for arbitrary $k \in \mathbb{N}, t \in \mathbb{R}$, but we have to pay 5 CHF each time.

For a general g , which is the degree of the best (i.e., highest-order) Taylor polynomial we can compute? How expensive will it be to compute it?

- (a) degree 3 and 20 CHF
- (b) degree 2 and 15 CHF
- (c) degree 2 and 10 CHF
- (d) degree 3 and 15 CHF

5.6. Polynomials. Let $P, Q \in \mathbb{R}[x_1, \dots, x_n]$ be polynomials of n variables, assume that there are positive numbers M, σ such that

$$|P(x) - Q(x)| \leq M|x|^\sigma \text{ for all } |x| \leq 1.$$

Show that P and Q have the same coefficients of order smaller than σ . That is, if we write

$$P(X) = \sum_{\alpha \in \mathbb{N}^n} a_\alpha X^\alpha, \quad Q(X) = \sum_{\alpha \in \mathbb{N}^n} b_\alpha X^\alpha,$$

then $a_\alpha = b_\alpha$ for all $|\alpha| < \sigma$.

5.7. Chain rule for a curve, or: a car on Earth. Let $R > 0$ and $\omega \in \mathbb{R}$ be constants. Let $\sigma : I \subset \mathbb{R} \rightarrow \mathbb{R}^2, \sigma(t) = (a(t), b(t))$, which we interpret as a curve in a “map” representing the longitude and latitude of a car at time t .

Define the “rotating” spherical coordinates $F : \mathbb{R}^2 \times \mathbb{R} \rightarrow \mathbb{R}^3$,

$$F((x, y), t) = (R \cos y \cos(\omega t + x), R \cos y \sin(\omega t + x), R \sin y),$$

representing what a point on the map looks like on Earth from a satellite (thus we consider the Earth as a rotating sphere with speed given by ω , and we forget about translation around the Sun) at time t .

Consider $r : \mathbb{R} \rightarrow \mathbb{R}^3$, $r(t) = F(\sigma(t), t)$, which is the position of the car at time t .

1. Compute the acceleration $r''(t)$ using the chain rule.
2. Assume that the acceleration is always proportional to the radial vector $r(t)$, i.e.

$$r''(t) = \lambda(t) r(t)$$

for some scalar function $\lambda(t)$. Show that a and b satisfy

$$b'' + \sin b \cos b (\omega + a')^2 = 0,$$

and

$$\cos b a'' - 2 \sin b (\omega + a') b' = 0.$$

Equivalently, away from the poles ($\cos b \neq 0$),

$$a'' - 2 \tan b (\omega + a') b' = 0.$$

Hints:

5.3 Revise Corollary 10.45.

5.4 In order to prove that a function **is** differentiable at 0 you have two tools

- the Definition of differentiability;
- the sufficient condition of Theorem 10.16.

In order to prove that a function **is not** differentiable at 0 you have two tools

- Show that the necessary condition of Proposition 10.14 does not hold;
- Show that for some C^1 curve $\gamma: (-\delta, \delta) \rightarrow \mathbb{R}^n$ the function $(f \circ \gamma)(t)$ is **not** differentiable at $t = 0$. This could be simpler as you have to study a function of only one variable.

5.4.3 These functions are C^∞ so try to compute them on $x + tv$ and find an expansion in powers of t ... If you find something it must be the Taylor polynomial by Corollary 10.45. From the Taylor polynomial it is immediate to read off the derivatives. For example

$$(x + tv)^2 = (x + tv)(x + tv) = x^2 + txv + tvx + t^2 = x^2 + (xv + vx)t + O(t^2)...$$

5. Solutions

Solution of 5.1:

(a) Solving the system

$$\begin{cases} 0 = \partial_x f = (1 - 2x^2)y^2 e^{-x^2-y^2} \\ 0 = \partial_y f = 2(y - y^3)x e^{-x^2-y^2} \end{cases}$$

we find that critical points are:

$$\{(s, 0) : s \in \mathbb{R}\}, \quad \left(\frac{1}{\sqrt{2}}, 1\right), \quad \left(-\frac{1}{\sqrt{2}}, 1\right), \quad \left(\frac{1}{\sqrt{2}}, -1\right), \quad \left(-\frac{1}{\sqrt{2}}, -1\right).$$

(b) Such a function cannot exist. By Proposition 10.14, $\partial_v g(x, y)$ must be a linear function of v when (x, y) are kept fixed, and this is not the case because of the term v_1^2 , which is nonlinear.

Solution of 5.2:

1. Since f is continuous and $\overline{U}$ is compact (closed and bounded), it must take within $\overline{U}$ its maximum and minimum value.
2. We are asked to find all $(x, y, z) \in U$ such that $\nabla f(x, y, z) = 0$, this is equivalent to solve the system

$$\begin{cases} 0 = \partial_x f = 4 - 2x \\ 0 = \partial_y f = -2y \\ 0 = \partial_z f = 0 \\ x^2 + y^2 + z^2 < 1 \end{cases}$$

which has no solutions since the first equation gives $x = 2$, contradicting the last condition. Not having critical points in U , by Proposition 11.3 we infer that f does not attain inside U its sup nor inf.

3. Since ∂U is compact, they must exist by the same reasoning of point 1.
4. Let's call $g(x, y, z) := x^2 + y^2 + z^2 - 1$. By Proposition 11.4, any critical point (relative to the minimization problem constrained to the boundary) solves the system

$$\begin{cases} 0 = \lambda_0 \partial_x f(x, y, z) + \lambda_1 \partial_x g(x, y, z) \\ 0 = \lambda_0 \partial_y f(x, y, z) + \lambda_1 \partial_y g(x, y, z) \\ 0 = \lambda_0 \partial_z f(x, y, z) + \lambda_1 \partial_z g(x, y, z) \\ \lambda_0^2 + \lambda_1^2 = 1 \\ g(x, y, z) = 0 \end{cases}$$

which becomes

$$\begin{cases} 0 = \lambda_0(4 - 2x) + 2\lambda_1x \\ 0 = \lambda_0(-2y) + 2\lambda_1y \\ 0 = 2\lambda_1z \\ \lambda_0^2 + \lambda_1^2 = 1 \\ x^2 + y^2 + z^2 = 1 \end{cases},$$

Notice that if $\lambda_0 = 0$ then we would get $x = y = z = 0$ which does not lie on ∂U , so setting $\lambda := \lambda_1/\lambda_0$ we have the equivalent system

$$\begin{cases} 0 = 2 - x + \lambda x = 2 + (\lambda - 1)x \\ 0 = -y + \lambda y = (\lambda - 1)y \\ 0 = \lambda z \\ x^2 + y^2 + z^2 = 1 \end{cases}$$

observe that if $\lambda = 0$ then we would have $x = -2$ which contradicts $(x, y, z) \in \partial U$. So we find $z = 0$ and the equivalent system:

$$\begin{cases} 0 = 2 + (\lambda - 1)x \\ 0 = (\lambda - 1)y \\ x^2 + y^2 = 1 \end{cases}$$

which forces $(x, \lambda) \in \{(-1, 3), (1, -1)\}$ and $y = 0$. Summing up the critical (x, y, z, λ) are

$$(1, 0, 0, -1) \quad (-1, 0, 0, 3).$$

5. Since we showed that $\max_{\bar{U}}$ is attained at some point $P \in \bar{U}$ and that $P \notin U$, we must have $P \in \partial U$ and, by Proposition 11.4, that P is critical for the constrained problem

$$\max_{\partial U} f.$$

Same reasoning for $\min_{\bar{U}} f$. So the only two candidates are

$$(1, 0, 0) \quad (-1, 0, 0).$$

We check:

$$f(1, 0, 0) = +4 \quad f(-1, 0, 0) = -4.$$

So the point $(1, 0, 0)$ is an absolute maximum for $f|_{\bar{U}}$ and the point $(-1, 0, 0)$ is an absolute minimum for $f|_{\bar{U}}$.

Solution of 5.3:

- (a) True. As $x \rightarrow 0$, we have $x_2x_1^2 = O(|x|^3) \leq O(|x|^2)$ and $O(x_1^2) \leq O(|x|^2)$.

- (b) False. The function $x_2x_1^2$ is **not** $O(x_1^3)$, in fact the ratio

$$\frac{x_2x_1^2}{x_1^3} = \frac{x_2}{x_1}$$

is unbounded in any neighbourhood of the origin.

- (c) True. Just apply the definition of differentiability with $Df(0) = 0$.
 (d) False. Take for example $f(x) = \max\{0, x_1\}^2$. Notice that saying that $f(x) = x_2x_1 + O(|x|^2)$, is exactly the same as saying that $f(x) = O(|x|^2)$.
 (e) False. This would be true if we had the *a priori* assumption that $f \in C^2$ in a small ball around the origin. Without this assumptions it is easy to construct counterexamples as in Analysis I, for example

$$f(x) = x_1x_2 + |x|^4 \sin(|x|^{-100}),$$

in such a way that $|\partial_1 f(x)| \rightarrow \infty$ as $|x| \rightarrow 0$, so $\partial_1 f(x)$ is not even continuous at $x = 0$, let alone differentiable.

Solution of 5.4:

1. These functions are differentiable away from $\{x_2 = 0\}$; the key point is that $f = g/h$ is differentiable where $h \neq 0$ and both g and h are differentiable. To compute the directional derivatives with respect to the direction vector $v = (v_1, v_2)$, we can then use the definition of directional derivative:

$$\partial_v f(x) = v \cdot \nabla f(x) = v_1 \partial_1 f(x) + v_2 \partial_2 f(x),$$

where $\nabla f(x)$ is the gradient of the function $f(x)$.

- $f(x) = \frac{x_1}{x_2}$. The gradient of f is $\nabla f(x) = \left(\frac{1}{x_2}, -\frac{x_1}{x_2^2}\right)$, so, the directional derivative is

$$\partial_v f(x) = \frac{v_1}{x_2} - \frac{v_2 x_1}{x_2^2}.$$

This function cannot be extended continuously to $\{x_2 = 0\}$; in particular, there is no differentiable extension either.

- $f(x) = e^{-\frac{x_1}{x_2}}$. The gradient of f is $\nabla f(x) = \left(-\frac{1}{x_2} e^{-\frac{x_1}{x_2}}, \frac{x_1}{x_2^2} e^{-\frac{x_1}{x_2}}\right)$, so, the directional derivative is

$$D_v f(x) = -\frac{v_1}{x_2} e^{-\frac{x_1}{x_2}} + \frac{v_2 x_1}{x_2^2} e^{-\frac{x_1}{x_2}}$$

This function cannot be extended continuously to $\{x_2 = 0\}$; in particular, there is no differentiable extension either.

2. These functions are differentiable away from $\{x = 0\}$, again by the quotient criterion.

- $f(x) = e^{x_1 x_2} \sin(x_1 + x_2^2)$. The gradient of f is

$$\nabla f(x) = \left(e^{x_1 x_2} [x_2 \sin(x_1 + x_2^2) + \cos(x_1 + x_2^2)], \right. \\ \left. e^{x_1 x_2} [x_1 \sin(x_1 + x_2^2) + 2x_2 \cos(x_1 + x_2^2)] \right).$$

So, the directional derivative is:

$$\partial_v f(x) = v_1 e^{x_1 x_2} [x_2 \sin(x_1 + x_2^2) + \cos(x_1 + x_2^2)] \\ + v_2 e^{x_1 x_2} [x_1 \sin(x_1 + x_2^2) + 2x_2 \cos(x_1 + x_2^2)]$$

- $f(x) = \frac{x_1^2}{x_1^2 + x_2^2}$. The gradient of f is

$$\nabla f(x) = \left(\frac{2x_1(x_1^2 + x_2^2) - 2x_1^3}{(x_1^2 + x_2^2)^2}, \frac{-2x_1^2 x_2}{(x_1^2 + x_2^2)^2} \right).$$

So, the directional derivative is:

$$\partial_v f(x) = v_1 \frac{2x_1 x_2^2}{(x_1^2 + x_2^2)^2} - v_2 \frac{2x_1^2 x_2}{(x_1^2 + x_2^2)^2}.$$

It has no continuous extension through the origin (observe that $\lim_{t \rightarrow 0, t \neq 0} f(t, 0) = 1$ but $\lim_{t \rightarrow 0, t \neq 0} f(0, t) = 0$).

- $f(x) = \frac{x_1^2 x_2}{x_1^2 + x_2^2}$. The gradient of f is

$$\nabla f(x) = \left(\frac{2x_1 x_2 (x_1^2 + x_2^2) - 2x_1^3 x_2}{(x_1^2 + x_2^2)^2}, \frac{x_1^2 (x_1^2 + x_2^2) - 2x_1^2 x_2^2}{(x_1^2 + x_2^2)^2} \right) \\ = \left(\frac{2x_1 x_2^3}{(x_1^2 + x_2^2)^2}, \frac{x_1^2 (x_1^2 - x_2^2)}{(x_1^2 + x_2^2)^2} \right).$$

So, the directional derivative is:

$$\partial_v f(x) = v_1 \frac{2x_1 x_2^3}{(x_1^2 + x_2^2)^2} + v_2 \frac{x_1^2 (x_1^2 - x_2^2)}{(x_1^2 + x_2^2)^2}.$$

It admits a continuous extension $\tilde{f}$, with $\tilde{f}(x) = f(x)$ for $x \neq 0$, through the origin, by putting $\tilde{f}(0) = 0$. We can see this since $|f(x)| = \frac{|x_1|^2 |x_2|}{|x|^2} \leq \frac{|x|^3}{|x|^2} = |x|$, which shows that $\lim_{x \rightarrow 0, x \neq 0} \tilde{f}(x) = 0$. On the other hand, $\tilde{f}$ is not differentiable at the origin: We have $\partial_{x_1} \tilde{f}(0)$ and $\partial_{x_2} \tilde{f}(0) = 0$, simply because $\tilde{f}(x) = 0$ if $x_1 = 0$ or $x_2 = 0$, but choosing $\nu = (1, 1)$ we have $\partial_\nu \tilde{f}(0) \neq 0 = \nu \cdot \nabla \tilde{f}$ (check!).

- $f(x) = \frac{x_1^2 x_2}{x_1^2 + x_2^4}$. The gradient of f is

$$\nabla f(x) = \left(\frac{2x_1 x_2 (x_1^2 + x_2^4) - 2x_1^3 x_2}{(x_1^2 + x_2^4)^2}, \frac{x_1^2 (x_1^2 + x_2^4) - 4x_1^2 x_2^3}{(x_1^2 + x_2^4)^2} \right) \\ = \left(\frac{2x_1 x_2^5}{(x_1^2 + x_2^4)^2}, \frac{x_1^2 (x_1^2 - 3x_2^4)}{(x_1^2 + x_2^4)^2} \right).$$

So, the directional derivative is:

$$\partial_v f(x) = v_1 \frac{2x_1 x_2^5}{(x_1^2 + x_2^4)^2} + v_2 \frac{x_1^2(x_1^2 - 3x_2^4)}{(x_1^2 + x_2^4)^2}.$$

It admits a continuous extension across the origin by putting $f(0,0) = 0$, since $|f(x)| \leq \frac{|x_1^2||x_2|}{|x_1|^2} = |x_2| \leq |x|$. On the other hand, the extension is not differentiable, which one checks as in the previous case.

3. We denote with $\langle \cdot, \cdot \rangle$ the standard scalar product of $\mathbb{R}^n$.

- $|x|$: Away from the origin, we compute

$$\partial_v |x| = \sum_{i=1}^n v_i \partial_i (\sqrt{|x|^2}) = \sum_{i=1}^n \frac{v_i}{2|x|} \partial_i (|x|^2) = \sum_{i=1}^n \frac{v_i x_i}{|x|} = \frac{\langle x, v \rangle}{|x|}.$$

It is **not** differentiable at $x = 0$ as can be seen by contradiction. Assume that for some $\xi \in \mathbb{R}^n$ we had

$$|x| = \langle x, \xi \rangle + o(|x|) \text{ as } |x| \rightarrow 0.$$

Then testing this along the lines $x = \pm t\xi$ for $t \downarrow 0$ we find that necessarily $\xi = 0$, so $|x| = o(|x|)$, which is impossible.

- $|x|^\alpha$: Away from the origin, we compute

$$\begin{aligned} \frac{\partial |x|^\alpha}{\partial v} &= \frac{\partial}{\partial v} (|x|^\alpha) = \frac{\partial}{\partial v} ((x_1^2 + x_2^2 + \dots + x_n^2)^{\alpha/2}) \\ &= \frac{\alpha}{2} (x_1^2 + x_2^2 + \dots + x_n^2)^{\alpha/2-1} \frac{\partial}{\partial v} (x_1^2 + x_2^2 + \dots + x_n^2) = \alpha |x|^{\alpha-2} \langle x, v \rangle \end{aligned}$$

For $\alpha > 1$ we can easily compute its partial derivatives at the origin too, obtaining 0 in any direction, and moreover we have

$$\lim_{|x| \rightarrow 0} \left| \frac{\partial |x|^\alpha}{\partial v} \right| = \lim_{|x| \rightarrow 0} O(|x|^{\alpha-1}) = 0.$$

By Theorem 10.16, $|x|^\alpha$ is then of class $C^1(\mathbb{R}^n)$. In the case $\alpha \leq 1$ we do not have differentiability at the origin.

- $\frac{x}{|x|}$: using Leibnitz and the previous identity

$$\partial_v (x/|x|) = \partial_v (x) |x|^{-1} + x \partial_v (|x|^{-1}) = \frac{v|x|^2 - \langle x, v \rangle x}{|x|^3}.$$

Observe that $x/|x|$ cannot be extended continuously at $x = 0$.

- $g(|x|)$: By chain rule

$$\partial_v g(|x|) = g'(|x|) \partial_v |x| = g'(|x|) \frac{\langle x, v \rangle}{|x|}$$

A reasoning similar to the first two above gives that g is differentiable at $x = 0$ if and only if $\lim_{t \rightarrow 0+} g'(t) = 0$.

- $x \cdot e$, being a linear function, it coincides with its differential. Let's check it:

$$\frac{\partial(x \cdot e)}{\partial v} = \frac{\partial}{\partial v}(x_1 e_1 + x_2 e_2 + \dots + x_n e_n) = e \cdot v.$$

- $g(x \cdot e)$: By the chain rule:

$$\frac{\partial g(x \cdot e)}{\partial v} = g'(x) \frac{\partial(x \cdot e)}{\partial v} = g'(x \cdot e)(e \cdot v)$$

and it is of class C^1 being a composition of C^1 functions.

4. The maps ax, a^T and $\text{Tr}(x)$ are linear so their differential is easily computed using the definition

$$\partial_v(ax) = av, \quad \partial_v(x^T) = v^T \text{ and } \partial_v(\text{Tr}(x)) = \text{Tr}(v).$$

For x^2 it is convenient to use Corollary 10.45 (it is of class C^2 since the entries of x^2 are polynomials of the entries of x).

$$(x + tv)^2 = (x + tv)(x + tv) = x^2 + txv + tvx + t^2 = x^2 + (xv + vx)t + O(t^2), \quad (1)$$

and so we must have

$$\partial_v x^2 = xv + vx \quad (\text{which is **not** } 2xv \text{ in general!!!}).$$

Similarly for x^3 one finds

$$\partial_v x^3 = x^2 v + xv x + vx^2.$$

If we apply the trace we can commute it with the derivative, so one can guess

$$\partial_v \text{Tr}(x^2) = \text{Tr}(xv + vx) = 2\text{Tr}(xv),$$

one can also check it tracing (1)

$$\text{Tr}((x + tv)^2) = \text{Tr}(x^2) + 2t\text{Tr}(xv) + t^2\text{Tr}(v^2) = \text{Tr}(x^2) + 2t\text{Tr}(xv) + O(t^2).$$

5. Define $U := \{x \in \mathbb{R}^{n \times n} : \det(x) \neq 0\}$ and notice that it is open (why?). We start observing that the function $\iota: x \mapsto x^{-1}$ is smooth (i.e., C^∞) from U in itself, indeed the formula for the entries of the inverse matrix is explicit and algebraic as long as we are in U .¹ Thus we know that there is some $\partial_v \iota(x) \in \mathbb{R}^{n \times n}$ such that

$$(x + tv)^{-1} = x^{-1} + t\partial_v \iota(x) + O(t^2), \quad \text{as } t \rightarrow 0.$$

¹That formula is complicated and we do not want to use it directly, but it is useful to know that it exists and that it is of the form

$$(x^{-1})_{ij} = \frac{\text{polynomial in the entries of } x}{\det x},$$

thus it is smooth as long as $\det(x) \neq 0$, that is to say $x \in U$.

We use again the algebraic properties of the inverse:

$$\begin{aligned}\mathbf{1}_{n \times n} &= (x + tv)^{-1}(x + tv) = (x + tv)(x^{-1} + t\partial_v \iota(x) + O(t^2)) \\ &= \mathbf{1}_{n \times n} + tvx^{-1} + tx\partial_v \iota(x) + O(t^2).\end{aligned}$$

Simplifying we find

$$vx^{-1} + x\partial_v \iota(x) = O(t),$$

so the only possibility is that

$$vx^{-1} + x\partial_v \iota(x) = 0,$$

which finally leads to

$$\partial_v \iota(x) = \partial_v(x^{-1}) = -x^{-1}vx^{-1}.$$

Now we can compute $\partial_v(x^{-2})$. We call $\sigma: U \rightarrow U$ the map $x \mapsto x^2$ and so that $x^{-2} = \sigma \circ \iota(x)$. Recall that we showed $\partial_w \sigma(y) = yw + wy$. We apply the chain rule

$$\partial_v(\sigma \circ \iota)(x) = \partial_{\partial_v \iota(x)} \sigma(\iota(x)) = \iota(x) \cdot \partial_v \iota(x) + \partial_v \iota(x) \cdot \iota(x) = -x^{-2}vx^{-1} - x^{-1}vx^{-2}.$$

Solution of 5.5: The answer is (a). Since f is given up to a $O(|x|^4)$ error, there is no hope to find any Taylor polynomial of degree larger than 3 (for example g could be the identity). Now Taylor expanding $(g \circ f)(x)$ we find

$$\begin{aligned}(g \circ f)(x) &= g(x_1 + x_1x_2 + O(|x|^4)) \\ &= g(0) + g'(0)(x_1 + x_1x_2 + O(|x|^4)) + \frac{g''(0)}{2}(x_1 + x_1x_2 + O(|x|^4))^2 \\ &\quad + \frac{g'''(0)}{6}(x_1 + x_1x_2 + O(|x|^4))^3 + O((x_1 + x_1x_2 + O(|x|^4))^4) \\ &= g(0) + g'(0)(x_1 + x_1x_2) + \frac{g''(0)}{2}(x_1^2 + 2x_1^2x_2) + \frac{g'''(0)}{6}x_1^3 + O(|x|^4).\end{aligned}$$

So we need the numbers $g(0), g'(0), g''(0), g'''(0)$ for which we need to pay 20 CHF. Notice that it would have been cheaper if we had the different assumption

$$f(x) = x_2x_1 + O(|x|^4) \text{ as } x \rightarrow 0.$$

Solution of 5.6: Without loss of generality, we can assume $Q = 0$ and that P has degree $N < \sigma$. Then we must have

$$\lim_{r \downarrow 0} \frac{|P(rx)|}{r^m} \leq \limsup_{r \downarrow 0} Mr^{\sigma-m} = 0 \text{ for all } m \leq N. \quad (2)$$

Assume that P is nonzero and take the shortest multiindex $\gamma \in \mathbb{N}^n$ such that there is $a_\gamma \neq 0$, so we have

$$\begin{aligned}|P(rx)| &\geq r^{|\gamma|} \left| \sum_{\alpha \in \mathbb{N}^n, |\alpha|=|\gamma|} a_\alpha x^\alpha \right| - r^{|\gamma|+1} \left| \sum_{\alpha \in \mathbb{N}^n, N \geq |\alpha| > |\gamma|} a_\alpha r^{|\alpha|-|\gamma|-1} x^\alpha \right| \\ &\geq r^{|\gamma|} \left| \sum_{\alpha \in \mathbb{N}^n, |\alpha|=|\gamma|} a_\alpha x^\alpha \right| - Cr^{|\gamma|+1}.\end{aligned}$$

Since $|\gamma| \leq N$, (2) entails $\left| \sum_{\alpha \in \mathbb{N}^n, |\alpha|=|\gamma|} a_\alpha x^\alpha \right| = 0$. But x was arbitrary and this means that all the a_α with $|\alpha| = |\gamma|$ are zero, which contradicts the choice of γ .

Solution of 5.7: Set

$$\theta(t) = \omega t + a(t), \quad \theta'(t) = \omega + a'(t), \quad \theta''(t) = a''(t).$$

Then

$$r(t) = (R \cos b \cos \theta, R \cos b \sin \theta, R \sin b).$$

It is convenient to introduce the orthonormal frame

$$\begin{aligned} e_r &= (\cos b \cos \theta, \cos b \sin \theta, \sin b), \\ e_b &= (-\sin b \cos \theta, -\sin b \sin \theta, \cos b), \\ e_\theta &= (-\sin \theta, \cos \theta, 0). \end{aligned}$$

Thus $r = R e_r$. Differentiating, one finds

$$\begin{aligned} e'_r &= b' e_b + \theta' \cos b e_\theta, \\ e'_b &= -b' e_r - \theta' \sin b e_\theta, \\ e'_\theta &= -\theta' \cos b e_r + \theta' \sin b e_b. \end{aligned}$$

Hence

$$r' = R e'_r = R(b' e_b + \theta' \cos b e_\theta).$$

Differentiating once more,

$$r'' = R \left(-(b'^2 + \theta'^2 \cos^2 b) e_r + (b'' + \theta'^2 \sin b \cos b) e_b + (\cos b \theta'' - 2 \sin b b' \theta') e_\theta \right).$$

Since $\theta' = \omega + a'$ and $\theta'' = a''$, this becomes

$$r'' = R \left(-(b'^2 + (\omega + a')^2 \cos^2 b) e_r + (b'' + \sin b \cos b (\omega + a')^2) e_b + (\cos b a'' - 2 \sin b (\omega + a') b') e_\theta \right).$$

This gives the acceleration required in part (1).

Now assume

$$r''(t) = \lambda(t) r(t)$$

for some scalar function λ . Since $r(t) = R e_r(t)$, the vector $r''(t)$ must have no components in the tangent directions e_b and e_θ . Therefore their coefficients vanish:

$$\begin{aligned} b'' + \sin b \cos b (\omega + a')^2 &= 0, \\ \cos b a'' - 2 \sin b (\omega + a') b' &= 0. \end{aligned}$$

Away from the poles, where $\cos b \neq 0$, the second equation is equivalently

$$a'' - 2 \tan b (\omega + a') b' = 0.$$

Therefore the equations satisfied by a and b are

$$\boxed{b'' + \sin b \cos b (\omega + a')^2 = 0, \quad \cos b a'' - 2 \sin b (\omega + a') b' = 0.}$$